

YEAR 10 SCIENCE CHEMICAL SCIENCES INVESTIGATION

RELATIVE REACTIVITY OF DIFFERENT METALS

Background

It is obvious that some metals are more reactive than others. For example, sodium metal reacts violently with water and its reactions with chlorine, steam and dilute hydrochloric acid solution are too explosive to even perform in the laboratory. On the other hand, silver metal does not react with any of these reagents. The metals can all be ranked in order of their reactivity in a list known as the **reactivity series of metals**. If a metal (A), that does not react with water, is placed into a solution that contains the ions of a different metal (B), then a reaction will only take place if metal A is more reactive than metal B.

Purpose

The aim of this investigation is to observe several reactions of a metal with the metal ions of other metals and then to list the given metals in their order of reactivity. You will need to complete the reactions in your group, but each person should then deduce the order of the metals on their own.

A suitable method is given below.

Equipment

Test tubes (x 6)

Samples of four different metals (zinc, magnesium, copper, and lead)

Solutions of four different metal ions ($\text{Zn}(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$, $\text{Cu}(\text{NO}_3)_2$, and $\text{Pb}(\text{NO}_3)_2$)

Method

1. Draw up a table to record your observations (if no reaction is observed write “no reaction”).

Solution Metal	$\text{Zn}(\text{NO}_3)_2$	$\text{Mg}(\text{NO}_3)_2$	$\text{Cu}(\text{NO}_3)_2$	$\text{Pb}(\text{NO}_3)_2$
Zinc	—			
Magnesium		—		
Copper			—	
Lead				—

- Place a piece of zinc metal into each of three test tubes.
- Add enough $\text{Mg}(\text{NO}_3)_2$ solution to one of the above test tubes to completely cover the zinc.
- Observe and note down any changes that may occur.
- Repeat steps 3 and 4, adding $\text{Cu}(\text{NO}_3)_2$ solution to the zinc in one of the other test tubes.
- Repeat steps 3 and 4, adding $\text{Pb}(\text{NO}_3)_2$ solution to the zinc in the final test tube.
- Repeat the entire procedure (steps 2 – 6) testing pieces of magnesium metal in three more test tubes and using the $\text{Zn}(\text{NO}_3)_2$, $\text{Cu}(\text{NO}_3)_2$, and $\text{Pb}(\text{NO}_3)_2$ solutions, respectively.
- Repeat the entire procedure (steps 2 – 6) testing pieces of copper metal with the $\text{Zn}(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$, and $\text{Pb}(\text{NO}_3)_2$ solutions, respectively.
- Repeat the entire procedure (steps 2 – 6) testing pieces of lead metal with the $\text{Zn}(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$, and $\text{Cu}(\text{NO}_3)_2$ solutions, respectively.
- List the four metals tested in this experiment in order of decreasing reactivity.
- Discuss and give clear explanations for the order of each metal in your list.
- Write a suitable conclusion for your experiment.

Marking Scheme

SECTION	MARK
Aim	1
Equipment	1
Method	1
Conducting - Safety glasses - Care with equipment - Careful observations - Laboratory attitude - Clean up at conclusion of experiment	5
Results Copy of table of observations	7
Metals listed correctly	4
Explanations for order of metals	4
Conclusion	2
TOTAL	

25	6F
23-24	5H
21-22	5M
19-20	5F
16-18	4H
13-15	4M
10-12	4F
7-9	3H
5-6	3M
3-4	3F
1-2	2H